

Hyphenation

A hyphen is used to show a joining or a connection. Meaning, function, and placement in the sentence are factors that determine whether a hyphen is needed.

Science style discourages the use of a solidus (/) for anything other than a ratio or division. A hyphen may be useful in place of a solidus, as serine-threonine protein kinase.

1) A hyphen is used to join various parts of speech when they function together (as an adjective) to modify a noun.

teacher-pupil interaction (noun with noun modifies a noun)
liquid-vapor interface
high-frequency oscillation (adverb with adjective before a noun)
very-low-mass star (adverb with adjective with noun) [check the style in the field]
very low density lipoprotein (adverb with adjective before a noun)
ice-cold water (adjective with adjective)
well-educated man (adverb with adjective)
3-year-old child or 100-million-year-old fossil (complex adjective to modify noun), but a 10 year old
calcium-binding compounds (noun with participle)
5-mg sample (unit of measurement to modify noun)

3-to-1 ratio (numerical adjective)

Care must be taken to avoid long strings of modifiers ("freight-train" modifiers). Reword rather than use hyphens.

When a descriptive phrase follows a noun, ordinarily it is not hyphenated:

the man was well educated
the child was 3 years old

When a unit modifier with “well” or “better” or the like is itself modified, it is not hyphenated (see *Chicago*). (adverb with adjective before a noun)

very well educated man [very refers to “well educated”]
but two well-studied microRNAs [two refers to microRNAs]

Verb phrases are often hyphenated

the sample was freeze-dried (per Webster's III, freeze-dry is hyphenated as a verb)

One must use care in deciding whether to insert a hyphen because it can change the meaning:

large vegetable crop yield (vegetable crop yield that is large)
large-vegetable crop yield (crop yield of large vegetables)
un-ionized versus unionized

2) Proper names and foreign words used adjectivally are not hyphenated, as:

New England winter
South African native
in situ measurements
German American ancestry (Do not hyphenate dual-heritage terms. Other examples: African American, Asian American, etc.)

but: Madame Schumann-Heink (composite surname)

3) Hyphens are used in chemical formulations and mathematical terminology:

carbon-14
non-hydrogen molecules (prefix to a chemical name)
C-6 (position in carbon chain)
1,2-dicyanobutane
p-tert-butylphenol
¹²⁵I-labeled
100- and 200-ml aliquots (suspended hyphens)
kilowatt-hour
three-dimensional (use 3D if used more than once)
 α -aminobutyric acid
GC-MS (after spelling out on first usage)
Twenty-one days later, he . . . (spell out numeral and unit at the beginning of a sentence if you can't rephrase) as Of the 31 instances, 20 were . . .

but: stainless steel rod
sodium chloride solution (do not hyphenate chemical names used as adjectives)
amino acid sequence
a axis (crystallography)
xy plane
x axis
cell surface receptor

4) We do not use hyphens in joining -ly adverbs to a modifier:

beautifully told story
highly objectionable viewpoint

but: scholarly-looking person (scholarly is an adjective, not an adverb)

5) Prefixes and suffixes are special cases. Under some circumstances, hyphens are required. Always consult a dictionary in deciding whether to hyphenate or close up.

Hyphens are called for in the following:

co-oligomerize	anti-American sentiment
co-worker	self-reliance
intra-atomic	mid-latitude
cross-sectional	sail-like
co-receptor	anti-apoptotic

but: antisocial	biennial
herself	cross section (n); cross-sectional (adj.)
posttranslationally	semiannual
intramural	worldwide
transocean	reexamination
catlike	overexposed

(see Table 6.1 from *Chicago Style Manual*, ed. 14)

To follow latest version of *Chicago* (ed. 17): Words with “-like” suffixes should only be closed if listed as such in Webster’s. Otherwise, these terms should be hyphenated. If hyphenated, compounds retain the hyphen both before and after a noun.

6) Use hyphens to distinguish between words spelled alike but different in meaning, as:

recover: to regain
but re-cover: to cover again
recount: to tell or relate
but re-count: to count a second time

7) Use hyphens in verb-preposition combinations used as modifiers:

the marked-down designer dress
a stepped-up campaign
trumped-up charges

but: the dress was marked down for the sale

8) We do use hyphens in comparatives and superlatives:

best-dressed woman
higher-frequency measurements
least-squares ratio
smaller-sized fragment

Hyphens should be used with comparatives or superlatives to ensure clarity if it is not clear from context:

(2 different meanings)

I have the most prized furs.

I have the most-prized furs.

- 9) Frequently, two-part items are hyphenated or not hyphenated depending on their common usage. Below are commonly agreed usages of such two-word items.

OPEN

α carbon
 α cells
 α chain
 α helix
 α polymer
 α strand

β barrel
 β cells
 β protein
 β sheet
 β subunit
 β amyloid

HYPHEN

α -amanitin
 α -antitrypsin
 α -bungarotoxin
 α -helical
 α -tubulin

β -adrenergic receptor
 β -amanitin
 β -globin
 β -hydroxyl group
 β -microglobulin
 β -tubulin

DO NOT hyphenate:

antibody binding specificity
DNA binding activity
DNA binding protein
receptor binding domain
T cell receptor
RNA binding protein

DO hyphenate:

ligand-binding domain
TRBP-binding protein

Editors and proofreaders should be aware that hyphenation is not always necessary, particularly when its use will unduly call attention to a term or actually change meaning.

aerosol specific gravity (specific gravity of an aerosol)
aerosol-specific gravity (not the same as above)

navy bean soup	(hyphens are not necessary in these cases because the terms used are common)
granule cell development	
dark blue area	
root mean square deviation	

Occasionally, editors will request that specific terms follow a designated style for their discipline.

Ex: spacetime
 waveform
 wavefront
 waveguide

BUT
 wave function
 wave packet
 wave vector

If a usage is unclear, the editor or author should be queried.

See also the list of spelling terms.

En Dash

We do not use an en dash in constructions such as

"from 10 – 20%" or "between \$50 – \$75."

We would use

"from 10 to 20%" and "between \$50 and \$75."

Science style calls for the use of an en dash in the following cases:

1) To indicate a range of reference numbers in the text or to indicate inclusive pages or a range of volumes or years in the references, as:

... (4–7), ... pp. 347–355, ... EPA Reports (1985–1988)

2) For an inclusive period, with consecutive years:

FY 1986–1987 winter of 1986–1987

3) In a two-word or two-part idea connected to another word or words to form an adjective:

non–blood-fed
thin plastic–reinforced tubing
Pst I–cleaved
HTLV-I–infected cell lines
specific receptor–binding site
insulin-like growth factor I–sensitive cells
cytochrome P-450–catalyzed formation
SDS–polyacrylamide gel electrophoresis
growth hormone–releasing factor
non–Fermi liquid-like
non–hydrogen bonding
20–amino acid sequence
amino acid–specific

Similarly with proper nouns:

New York–Boston train, Hoffman–La Roche, van't Hoff–Le Bel

4) When an acronym is first spelled out and is used adjectivally:

adult T cell leukemia (ATL)–derived, **but** ATL-derived

5) To indicate a representative's or senator's party affiliation and state:

Senator Carl Levin (D–MI)

6) In tables (but not in text) to indicate a range of numbers:

Ages	17–34	46–52
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7) In referring to a line, as line A–A'.

8) To designate a single bond, as C–H (but use a hyphen when elements are spelled out; e.g., carbon-hydrogen bond).

9) Between restriction endonucleases:

PW III–NCO I fragment

We usually use an en dash for a fragment cut by two restriction endonucleases as Bam HI–Eco RI, but we use a slant line when two restriction endonucleases cleave a sequence at the same spot, such as Eco RI/Bam HI.

Em Dash

We use sparingly.

1) Use em dashes to set off words that would otherwise be misread:

Incorrect

All three experimental parameters, temperature, time, and concentration, were strictly followed.

Correct

All three experimental parameters—temperature, time, and concentration—were strictly followed.

2) Do not use em dashes to separate phrases or nonrestrictive clauses if another form of punctuation can be used:

Incorrect

Jones—not Riesberg—obtained good correlation of results and calculations.

Correct

Jones, not Riesberg, obtained good correlation of results and calculations.

Incorrect

The singly charged complexes—which constituted bands 1 and 3—liberated maleate anion upon decomposition.

Correct

The singly charged complexes, which constituted bands 1 and 3, liberated maleate anion upon decomposition.

See also *Chicago*, ed. 14, sections 5.106 to 5.114

3-Em Dash

We no longer use 3-em dashes.